

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV (NEP) Examination, 2025-26

CHEMISTRY**Paper Code: CHEMN41 : Physical Chemistry-II**

Time: 2.5 Hours

Maximum Marks: 70

REF NO. Conf/Even/N/2025-26/08

*Instructions: Answer **five** questions including Question No. 1 which is compulsory. All questions carry equal marks (14 marks each).*

Question 1. (Compulsory, 14 Marks Total)

- (a) Define the quantum yield of a photochemical reaction. [2]
- (b) What do you mean by fugacity? [2]
- (c) Write the rate equation of a unimolecular surface reaction, identify the terms involved and provide one example. [2]
- (d) Write the law of photochemical equivalence and as per this law, what should be the quantum yield? [2]
- (e) What are the general characteristics of radioactive decay? [2]
- (f) How will you synthesise radioactive sodium in the laboratory? [2]
- (g) What is artificial radioactivity? Give an example. [2]

Question 2.

- (a) Write the expression to find out the Gibbs energy of mixing (ΔG_{mix}) and the entropy of mixing (ΔS_{mix}). [3]
- (b) Write van't Hoff equation for Osmotic pressure and discuss how the molecular weight of a polymer can be measured through osmotic pressure measurement? [4]
- (c) Derive Gibbs-Duhem equation and explain the significance of the Gibbs-Duhem equation in multi-component systems. [7]

Question 3.

- (a) Write any two deactivation processes of a photochemically activated molecule. [3]
- (b) What do you mean by primary and secondary processes of photochemical reaction? And mention the process responsible for the variation of the quantum yield. [4]

- (c) Write the Brunauer-Emmett-Teller (BET) adsorption isotherm equation. Using the linear form of the BET equation, explain how it can be used to determine the specific surface area of the solid. [7]
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Question 4.

- (a) What are the two differences between thermal equilibrium and photostationary state? [3]
- (b) What is Michaelis-Menten equation? Draw the Lineweaver-Burk plot (double reciprocal plot) and write the values of the slope and intercept of the plot. [4]
- (c) Discuss the fluorescence quenching, considering the dimerization of anthracene and derive the Stern-Volmer equation. [7]
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Question 5.

- (a) Describe the Duhem-Margules equation with one of its applications. [3]
- (b) Depicting the graph, discuss the influence of the pH and the temperature on the enzyme catalysed reaction. [4]
- (c) Determine the rate of photochemical decomposition of HI (showing the reaction mechanism) and the quantum yield of this process. [7]
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Question 6.

- (a) Describe the concept of Bohr's compound nucleus theory of nuclear reactions. [3]
- (b) What are radioisotopes? Write the precautions to be taken in the use of radioisotopes as tracer. [4]
- (c) Derive an expression for the decay constant of a radioactive substance. How is this decay constant related to half-life? [7]
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Question 7.

- (a) What are the different types of nuclear reactions? Describe radiative capture and transfer reactions with suitable examples. [6]
- (b) Explain the direct isotopic dilution analysis and inverse isotopic dilution analysis methods. [8]
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Question 8. Write short notes on any **two** of the following: [2 × 7 = 14]

- (a) Osmosis and osmotic pressure
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- (b) Compton scattering
- (c) Photochemical reactions between H_2 and Br_2 and derive thereof the rate of formation of HBr .